3.3.2 Vieta's formula for Polynomial Equations

What we have learnt for quadratic polynomial, can be extended to polynomials of higher degree. In this section we study the relations of the zeros of a polynomial of higher degree with its coefficients. We also learn how to form polynomials of higher degree when some information about the zeros are known. In this chapter, we use either zeros of a polynomial of degree n or roots of polynomial equation of degree n .

3.3.2 (a) The Fundamental Theorem of Algebra

If a is a root of a polynomial equation $P(x)=0$, then $(x-a)$ is a **factor** of $P(x)$. So, $\deg(P(x))\geq 1$. If a and b are roots of $P(x)=0$ then $(x-a)(x-b)$ is a factor of $P(x)$ and hence deg $(P(x))\geq 2$. Similarly if $P(x)=0$ has n roots, then its degree must be greater than or equal to n. In other words, a **polynomial equation of degree n cannot have more than n roots**.

In earlier classes we have learnt about "multiplicity". Let us recall what we mean by "multiplicity". We know if $(x-a)^k$ is a factor of a polynomial equation $P(x)=0$ and $(x-a)^{k+1}$ is not a factor of the polynomial equation, $P(x)=0$, then a is called a root of **multiplicity** k . For instance, 3 is a root of multiplicity 2 for the equation $x^2-6x+9=0$ and $x^3-7x^2+159x-9=0$. Though we are not going to use complex numbers as coefficients, it is worthwhile to mention that the imaginary number $2+i$ is a root of multiplicity 2 for the polynomials $x^2-(4+2i)x+3+4i=0$ and $x^4-8x^3+26x^2-40x+25=0$. If a is a root of multiplicity 1 for a polynomial equation, then a is called a **simple root** of the polynomial equation.

If $P(x)=0$ has n roots counted with multiplicity, then also, we see that its degree must be greater than or equal to n. In other words, "a polynomial equation of degree n cannot have more than n roots, even if the roots are counted with their multiplicities".

One of the important theorems in the theory of equations is the fundamental theorem of algebra. As the proof is beyond the scope of the Course, we state it without proof.

Theorem 3.1 (The Fundamental Theorem of Algebra)
Every polynomial equation of degree $n\geq 1$ has at least one root in $\mathbb{C}$.

Using this, we can prove that a polynomial equation of degree n has at least n roots in $\mathbb{C}$ when the roots are counted with their multiplicities. This statement together with our discussion above says that

a polynomial equation of degree n has exactly n roots in $\mathbb{C}$ when the roots are counted with their multiplicities.

Some authors state this statement as the fundamental theorem of algebra.

3.3.2(b) Vieta's Formula

(i) Vieta's Formula for Polynomial equation of degree 3

Now we obtain these types of relations to higher degree polynomials. Let us consider a general cubic equation

$$ax^3+bx^2+cx+d=0.$$

By the fundamental theorem of algebra, it has three roots. Let α , β , and γ be the roots. Thus we have

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